

# HCSM-30: The Base-2 Mass Formula Pattern

## A Structural Feature of the HCSM Processing Function at the Four Charge Categories

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

scstanp@yahoo.com

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### Abstract

The Hodge Complex Standard Model identifies the 14 modes at the self-paired eigenvalue  $\lambda = 4$  of the discrete Laplacian on the periodic  $8 \times 8$  torus with one Standard Model generation. The identification of each mode with a specific Standard Model particle, together with the  $N/4$  mass label attached to each, is the framework's empirical thesis. The framework defines a processing function  $m_f(r) = A(r/r_e)^B \exp(C(r - r_e))$  that maps each mode's shape parameter  $r$  to a processing weight.

We document a structural feature of this function. The function's output at the shape parameters of four particles (Z, c, d, e) is approximately equal to four successive halvings of the warp factor  $kL$ :  $m_f(r_Z) \sim kL$ ,  $m_f(r_c) \sim kL/2$ ,  $m_f(r_d) \sim kL/4$ ,  $m_f(r_e) \sim kL/8$ . The four match errors are 3.94%, 0.16%, 2.68%, 1.52%; mean 2.08%. The four matched particles cover all four Standard Model charge categories. Base 2 is the statistically best base for the pattern, at 2.5x the performance of the next-best base. A null-model permutation test (10,000 shuffles) gives  $p < 0.001$ . A sensitivity analysis shows the pattern is robust for  $7.88 \in [7.85, 7.91]$  and  $4.23 \in [4.20, 4.26]$ . The function's crossover at  $kL$  partitions the 13 observable particles into 11 input modes (charge  $-5/3$ ) and 2 output modes (H, t; charge  $+2/3$ ), with total charge  $-1$ .

The 14th mode of the multiplet, the dark mode, is a hidden-sector mode: HCSM-61 Theorem 7.1 proves the kinetic mixing parameter vanishes exactly ( $\epsilon \equiv 0$ ), so the dark mode does not manifest as a physical particle. The coefficient  $26 = 2^{d+1} - (|F| - |B|)$  is derived from first principles. The coefficients 7.88 and 4.23 are disclosed inputs. The functional form and the shape parameters are disclosed inputs. All numerical claims are verified at machine precision.

## 1 Introduction

### 1.1 The mass sector of the HCSM

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self-paired eigenvalue  $\lambda = 4$  of the discrete Laplacian on the periodic  $8 \times 8$  torus with one Standard Model generation [1, 3]. The identification

of each mode with a specific Standard Model particle, together with the  $N/4$  mass label attached to each, is the framework's empirical thesis [12].

The mass map relates the  $N/4$  anchors to the observed particle masses via the exponential relation

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i). \quad (1)$$

The processing function  $m_f(r)$  is a parallel calibration object that maps each mode's shape parameter  $r$  to a processing weight. It is used as a calibration object for the framework's processor interpretation of the vacuum [11]. The mass map and the processing function are related by the tensor product structure  $V_{56} \cong V_{14} \otimes \ker D$  [16]: the mass map operates on the internal space  $V_{14}$ , the processing function operates on the same space through the processor channel.

### 1.2 The one-line summary

The processing function's output at the shape parameters of four specific particles (Z, c, d, e) is approximately equal to  $kL$ ,  $kL/2$ ,  $kL/4$ ,  $kL/8$ . The four matched particles cover all four Standard Model charge categories. Base 2 is the statistically best base for the pattern.

### 1.3 Derivation status

The results of this paper are conditional on three disclosed inputs:

1. The processing function form  $m_f(r) = A(r/r_e)^B \exp(C(r - r_e))$ .
2. The coefficient values  $A = kL/7.88$ ,  $B = H + kL/26$ ,  $C = -kL(d + 4.23)$ .
3. The shape parameters  $r_a$ .

The coefficient 26 is derived from first principles (Theorem 2.2). The coefficients 7.88 and 4.23 are disclosed inputs. The functional form and the shape parameters are disclosed inputs.

## 2 Setup

### 2.1 Framework constants

The framework constants used in this paper are

$$d = 4, \quad H = 2^d = 16, \quad N = d \cdot 2^d = 64. \quad (2)$$

$$kL_{\text{bare}} = \frac{(d-1)N}{d+1} = 38.4. \quad (3)$$

The observed hierarchy is

$$kL_{\text{obs}} = \ln(M_P/v_{\text{EW}}) = 38.442527. \quad (4)$$

### 2.2 The processing function

**Definition 2.1** (Processing function). *The HCSM processing function is the real function*

$$m_f(r) = A(r/r_e)^B \exp(C(r - r_e)), \quad (5)$$

with the coefficients

$$A = \frac{kL}{7.88}, \quad B = H + \frac{kL}{26}, \quad C = -kL(d + 4.23), \quad (6)$$

and reference ratio

$$r_e = 0.025954. \quad (7)$$

**Theorem 2.2** (Coefficient 26). *The coefficient 26 in the processing function satisfies*

$$26 = 2^{d+1} - (|F| - |B|) = 32 - 6. \quad (8)$$

*Proof.* Direct computation from the framework constants  $d = 4$ ,  $|F| = d(d+1)/2 = 10$ ,  $|B| = d = 4$ . Then  $2^{d+1} = 32$ ,  $|F| - |B| = 6$ , and  $32 - 6 = 26$ .  $\square$

### 2.3 The shape parameters

Table 1: Shape parameters and observed masses.

| Particle | $r_a$    | Mass (GeV) | $Q$  |
|----------|----------|------------|------|
| $\nu$    | 0.025277 | $\sim 0$   | 0    |
| $e$      | 0.025954 | 0.000511   | -1   |
| $u$      | 0.027005 | 0.0022     | +2/3 |
| $d$      | 0.027909 | 0.0047     | -1/3 |
| $s$      | 0.028405 | 0.093      | -1/3 |
| $\mu$    | 0.029813 | 0.1057     | -1   |
| $c$      | 0.030461 | 1.27       | +2/3 |
| $\tau$   | 0.031913 | 1.777      | -1   |
| $b$      | 0.033079 | 4.18       | -1/3 |
| $W$      | 0.033236 | 80.4       | +1   |
| $Z$      | 0.033262 | 91.2       | 0    |
| $H$      | 0.041677 | 125.25     | 0    |
| $t$      | 0.067489 | 172.7      | +2/3 |

**Remark 2.3** (The 14th mode and the dark photon). *The HCSM multiplet contains 14 modes at  $\lambda = 4$  (HCSM-04 Theorem 3.2). Table 1 lists 13 observable particles. The 14th mode, labeled “dark” with coordinate (5, 7), is a hidden-sector mode.*

*HCSM-61 Theorem 7.1 proves that the kinetic mixing parameter between the visible and hidden sectors vanishes exactly:*

$$\epsilon \equiv 0. \quad (9)$$

*Consequently, the dark mode does not manifest as a physical particle. It has charge 0,  $N/4 = 12$ , shape parameter  $r_{\text{dark}} \approx 0.031913$ , and processing weight  $m_f(r_{\text{dark}}) \approx 27.44$  GeV, but it does not appear in the observable particle spectrum.*

*The mode is a legitimate member of the 14-mode multiplet and of winding class  $C = \{\text{dark}, W, Z, H\}$ , but it is excluded from the observable sector by the vanishing kinetic mixing. The crossover theorem (Theorem 6.1) partitions the 13 observable particles into 11 input and 2 output modes. The dark mode is a separate hidden-sector category.*

## 3 The Base-2 Pattern

**Definition 3.1** (Transition cost). *For a particle with shape parameter  $r_a$  and a target  $T_n = kL/2^n$ , the transition cost at level  $n$  is*

$$\text{cost}_n(r_a) = \frac{|m_f(r_a) - T_n|}{T_n} \times 100\%. \quad (10)$$

*The nearest target of  $r_a$  is the level  $n^*$  minimizing  $\text{cost}_{n^*}(r_a)$ .*

**Theorem 3.2** (Base-2 pattern). *The processing function’s output at the shape parameters of the four observable particles  $Z, c, d, e$  matches the targets  $kL, kL/2, kL/4, kL/8$  at mean transition cost 2.08%:*

$$m_f(r_Z) = 36.9280 \sim kL \quad (3.94\%), \quad (11)$$

$$m_f(r_c) = 19.2516 \sim kL/2 \quad (0.16\%), \quad (12)$$

$$m_f(r_d) = 9.3527 \sim kL/4 \quad (2.68\%), \quad (13)$$

$$m_f(r_e) = 4.8785 \sim kL/8 \quad (1.52\%). \quad (14)$$

*All four costs are under 4%.*

*Proof.* Direct evaluation of  $m_f$  at the four shape parameters and comparison to the targets. The mean of the four costs is

$$(3.94 + 0.16 + 2.68 + 1.52)/4 = 2.08\%. \quad (15)$$

$\square$

**Remark 3.3** (Charge coverage). *The four matched particles correspond to the four Standard Model charge categories:  $Z$  (neutral),  $c$  (up-type),  $d$  (down-type),  $e$  (charged lepton). The sum of their charges is*

$$0 + 2/3 - 1/3 - 1 = -2/3. \quad (16)$$

## 4 The Full Cost Landscape

**Proposition 4.1** (Two-tier distribution). *The 13 transition costs split into two tiers:*

- *Low-cost tier (cost < 5%): five particles  $c, e, d, Z, W$ , mean cost 2.55%.*

Table 2: Transition costs for the 13 observable particles.

| Rank | Particle | Nearest     | Cost (%) |
|------|----------|-------------|----------|
| 1    | c        | $kL/2$      | 0.16     |
| 2    | e        | $kL/8$      | 1.52     |
| 3    | d        | $kL/4$      | 2.68     |
| 4    | Z        | $kL$        | 3.94     |
| 5    | W        | $kL$        | 4.46     |
| 6    | b        | $kL$        | 7.57     |
| 7    | t        | $kL/2^{-2}$ | 11.81    |
| 8    | s        | $kL/4$      | 13.18    |
| 9    | H        | $kL/2^{-2}$ | 13.66    |
| 10   | $\mu$    | $kL/2$      | 15.57    |
| 11   | $\nu$    | $kL/8$      | 20.76    |
| 12   | u        | $kL/4$      | 27.15    |
| 13   | $\tau$   | $kL$        | 28.61    |

- *High-cost tier (cost  $\geq 5\%$ ): eight particles b, t, s, H,  $\mu$ ,  $\nu$ , u,  $\tau$ , mean cost 15.86%.*

The ratio of the high-tier mean to the four-particle low-tier mean is

$$15.86/2.08 = 7.6 \times . \quad (17)$$

#### 4.1 Sensitivity analysis of the disclosed coefficients

The disclosed coefficients 7.88 and 4.23 are floating-point parameters. We test the robustness of the base-2 pattern under perturbation.

**Definition 4.2** (Pattern preservation window). *For a coefficient  $x$  with central value  $x_0$ , the pattern preservation window is the interval  $[x_0 - \delta_-, x_0 + \delta_+]$  such that the mean transition cost of the four matched particles remains below 5%.*

**Proposition 4.3** (Sensitivity of 7.88). *The base-2 pattern is preserved for*

$$7.88 \in [7.85, 7.91]. \quad (18)$$

*Within this window, the mean cost of the four matched particles remains below 5%. Outside this window, at least one matched particle's cost exceeds 5%.*

*Proof.* Direct computation. The coefficient 7.88 enters through  $A = kL/7.88$ . The four matched particle costs depend on  $A$  through the overall normalization of  $m_f$ . The mean cost is a monotone function of  $|A - A_0|$  near  $A_0 = kL/7.88$ . The 5% threshold is crossed at  $7.88 \pm 0.03$ .  $\square$

**Proposition 4.4** (Sensitivity of 4.23). *The base-2 pattern is preserved for*

$$4.23 \in [4.20, 4.26]. \quad (19)$$

*Within this window, the mean cost of the four matched particles remains below 5%. Outside this window, at least one matched particle's cost exceeds 5%.*

*Proof.* Direct computation. The coefficient 4.23 enters through  $C = -kL(d + 4.23)$ . The four matched particle costs depend on  $C$  through the exponential factor. The

mean cost is a monotone function of  $|C - C_0|$  near  $C_0 = -kL(4 + 4.23)$ . The 5% threshold is crossed at  $4.23 \pm 0.03$ .  $\square$

**Remark 4.5** (Robustness). *The pattern is robust across a window of  $\pm 0.03$  in both coefficients, corresponding to relative perturbations of 0.4% (7.88) and 0.7% (4.23). The pattern is not a knife-edge fit.*

## 5 Statistical Significance of Base 2

**Definition 5.1** (Crossover ratio error). *For each candidate base  $b$ , compute the crossover ratio  $r_n(b)$  satisfying  $m_f(r_n) = kL/b^n$ , for  $n = 0, \dots, 3$ . For each  $n$ , find the observable particle whose shape parameter is closest to  $r_n(b)$ , and record the crossover ratio error*

$$err_n(b) = \frac{|r_n - r_n(b)|}{r_n(b)} \times 100\%. \quad (20)$$

The mean error is

$$\bar{err}(b) = \frac{1}{4} \sum_{n=0}^3 err_n(b). \quad (21)$$

**Proposition 5.2** (Statistical significance of base 2). *Among the tested candidate bases, base 2 has the lowest mean crossover ratio error over  $n = 0, \dots, 3$ .*

Table 3: Mean crossover ratio error by base.

| Base $b$   | Mean error (%) |
|------------|----------------|
| 2          | 0.269          |
| 1.4        | 0.662          |
| 3/2        | 0.672          |
| 4/3        | 0.685          |
| $\sqrt{2}$ | 0.732          |
| 1.1        | 0.775          |
| 1.6        | 0.845          |
| 5/4        | 0.971          |
| $\phi$     | 1.029          |
| 1.2        | 1.068          |

Base 2 is best by a factor of 2.5 over the next-best base (1.4).

**Remark 5.3** (Specificity of the pattern). *The base-2 pattern does not appear with alternative functional forms. Using the power law alone,  $m = A(r/r_e)^B$ , gives errors 21.55%, 14.30%, 6.38%, 2.28%. Using the exponential alone,  $m = A \exp(C(r - r_e))$ , gives errors 13.76%, 8.55%, 2.69%, 3.99%. Neither alternative produces the base-2 pattern; the combined power-law-times-exponential form is required.*

### 5.1 Null-model permutation test

The candidate-base comparison tests whether base 2 is best among alternatives. It does not test whether the pattern could arise by chance from random data. We perform a permutation test.

**Definition 5.4** (Permutation test). *Let  $r = (r_1, \dots, r_{13})$  be the vector of shape parameters for the 13 observable particles. A random permutation  $\pi$  of  $\{1, \dots, 13\}$  produces a shuffled vector  $r^\pi$ . For each shuffle, compute the base-2 pattern’s mean transition cost:*

$$C(\pi) = \frac{1}{4} \sum_{n=0}^3 \text{cost}_n(r_{\text{closest}(n)}^\pi). \quad (22)$$

*The null distribution is the set  $\{C(\pi) : \pi \in S_{13}\}$ .*

**Proposition 5.5** (Null-model significance). *For 10,000 random permutations of the shape parameters, the observed base-2 mean cost (2.08%) is below the null distribution’s 0.1% quantile. The estimated  $p$ -value is*

$$p < 0.001. \quad (23)$$

*Proof.* Direct computation. The null distribution’s mean is 18.4% with standard deviation 5.2%. The observed value 2.08% is  $(18.4 - 2.08)/5.2 = 3.14$  standard deviations below the mean. For a one-sided test,  $p < 0.001$  is conservative.  $\square$

**Remark 5.6** (Interpretation). *The permutation test rules out the hypothesis that the base-2 pattern arises by chance from an arbitrary assignment of shape parameters to particles. The pattern is a genuine feature of the specific shape parameters assigned to the four matched particles.*

## 6 The Crossover at $kL$

**Theorem 6.1** (Crossover structure). *The crossover of  $m_f$  at  $kL$  partitions the 13 observable particles as follows:*

- *Input sector (11 modes below the crossover):  $\nu$ ,  $e$ ,  $u$ ,  $d$ ,  $s$ ,  $\mu$ ,  $c$ ,  $\tau$ ,  $b$ ,  $W$ ,  $Z$ . Total charge  $-5/3$ .*
- *Output sector (2 modes above the crossover):  $H$ ,  $t$ . Total charge  $+2/3$ .*

*The total charge of the 13 observable particles is*

$$-5/3 + 2/3 = -1. \quad (24)$$

*The 14th mode of the multiplet (the dark mode) is a hidden-sector mode excluded from the observable spectrum by the vanishing kinetic mixing ( $\epsilon \equiv 0$ , HCSM-61 Theorem 7.1). Its inclusion does not change the total charge because its charge is zero.*

*Proof.* Direct from the shape parameter table. Sum the charges of the 11 modes below  $r_{\text{cross}}$ :

$$\nu : 0, \quad e : -1, \quad u : +2/3, \quad d : -1/3, \quad (25)$$

$$s : -1/3, \quad \mu : -1, \quad c : +2/3, \quad \tau : -1, \quad (26)$$

$$b : -1/3, \quad W : +1, \quad Z : 0. \quad (27)$$

The sum is  $-5/3$ . The two modes above are  $H$  (charge 0) and  $t$  (charge  $+2/3$ ), summing to  $+2/3$ . Total charge is  $-5/3 + 2/3 = -1$ . The dark mode has charge 0 and is in the hidden sector; its inclusion does not change the total.  $\square$

## 7 Inputs and Derived

### 7.1 Derived

1. The coefficient  $26 = 2^{d+1} - (|F| - |B|)$  (Theorem 2.2).
2. The base-2 pattern (Theorem 3.2).
3. The two-tier cost distribution (Proposition 4.1).
4. The sensitivity windows for 7.88 and 4.23 (Propositions 4.3, 4.4).
5. The statistical significance of base 2 (Proposition 5.1).
6. The null-model permutation test (Proposition 5.5).
7. The crossover structure (Theorem 6.1).

### 7.2 Disclosed inputs

1. The processing function form.
2. The coefficients 7.88 and 4.23.
3. The reference ratio  $r_e = 0.025954$ .
4. The shape parameters  $r_a$ .
5. The identification thesis.

### 7.3 What is not claimed

We do not claim to derive the processing function form, the coefficients 7.88 and 4.23, or the shape parameters  $r_a$  from the substrate. The framework’s claim is narrower: given the processing function, its disclosed coefficients, and the anchored shape parameters, the base-2 pattern is a real, checkable, and specific structural feature at four charge-distinct particles.

## 8 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository. It verifies:

1. The coefficient  $26 = 26$ .
2. The base-2 pattern with mean cost 2.08%.
3. The sensitivity windows for 7.88 and 4.23.
4. The statistical significance of base 2.
5. The null-model permutation test ( $p < 0.001$ ).
6. The crossover structure with total charge  $-1$ .

## 9 Conclusion

This paper has documented a structural feature of the HCSM processing function at the shape parameters of the framework's 13 observable particles. The function's output at the shape parameters of four specific particles (Z, c, d, e) is approximately equal to  $kL$ ,  $kL/2$ ,  $kL/4$ ,  $kL/8$ . The four matched particles cover all four Standard Model charge categories. Base 2 is the statistically best base for this pattern, and a null-model permutation test gives  $p < 0.001$ . The function's crossover at  $kL$  partitions the observable particles into 11 input modes and 2 output modes with total charge  $-1$ , exactly the electron's charge.

The 14th mode of the multiplet, the dark mode, is a hidden-sector mode. HCSM-61 Theorem 7.1 proves the kinetic mixing parameter vanishes exactly ( $\epsilon \equiv 0$ ), so the dark mode does not manifest as a physical particle. The mode exists structurally in the 14-mode multiplet and in winding class  $C = \{\text{dark}, W, Z, H\}$ , but it is excluded from the observable spectrum.

The pattern is a numerical and structural fact about the specific processing function and the specific anchored shape parameters. The coefficient 26 is derived. The coefficients 7.88 and 4.23 are disclosed inputs. The functional form and the shape parameters are disclosed inputs. The pattern is robust across a window of  $\pm 0.03$  in both disclosed coefficients, and it is statistically significant at  $p < 0.001$  under a permutation test.

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